

CS-736 Assignment: Statistical Image Modeling, Image Denoising

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1. (25 points) Bayesian Denoising of a Phantom Magnetic Resonance Image.

Get the 2D noiseless image and the 2D noisy image in the “data” folder.

Implement a **maximum-a-posteriori Bayesian image-denoising** algorithm that uses a suitable noise model (i.i.d. Gaussian or independent Rician) coupled with a MRF prior that uses a 4-neighbor neighborhood system (each pixel has 4 neighbors: left, right, up, down; the neighborhood wraps around at image boundaries) that has cliques of size *no* more than 2.

Use gradient ascent optimization with dynamic step size; implementing a fixed step size isn't acceptable. Ensure that the value of the objective function, i.e., the log posterior, at each iteration increases. Use the noisy image as the initial estimate.

Use 3 different MRF priors where the potential functions $V(x_i, x_j) := g(x_i - x_j)$ underlying the MRF penalize the difference between the neighboring voxel values x_i, x_j as follows (see class notes for details). You may rely on the “circshift()” function in Matlab when computing differences between every pixel in the image and its neighbors.

Introduce real-valued parameters $\alpha > 0$ and $\beta > 0$ to control the weighting between the prior and the likelihood.

Specifically, implement the following functionality as part of the denoising algorithm:

(a) (2 points)

A suitable noise model.

(b) (2 points)

MRF prior: Quadratic function: $g_1(u) := |u|^2$.

(c) (2 points)

MRF prior: Discontinuity-adaptive Huber function: $g_2(u) := 0.5|u|^2$, when $|u| \leq \gamma$ and $g(u) := \gamma|u| - 0.5\gamma^2$, when $|u| > \gamma$, where $0 < \gamma < \infty$ is a constant.

(d) (2 points)

MRF prior: Discontinuity-adaptive function: $g_3(u) := \gamma|u| - \gamma^2 \log(1 + |u|/\gamma)$, where $0 < \gamma < \infty$ is a constant.

For each MRF prior, manually tune the parameters to denoising the noisy image in order to achieve the least possible relative root-mean-squared error (RRMSE). The RRMSE for 2 images A and B is defined as :

$$\text{RRMSE}(A, B) = \sqrt{\sum_p (|A(p)| - |B(p)|)^2} / \sqrt{\sum_p |A(p)|^2}$$
, where the summation is over all pixels p ; always use the noiseless image as A .

Report the following:

(a) (0 point)

Report the RRMSE between the noisy and noiseless images.

(b) (8 points)

Report the optimal values of the parameters and the corresponding RRMSEs for each of the 3 denoising algorithms. For each optimal parameter value reported (for each of the 3 denoising algorithms), give evidence of the optimality of the reported values by reporting the RRMSE values for two nearby parameter values (around the optimal) at plus/minus 20% of the optimal value. That is, if a^*, b^* are the optimal parameter values, then report:

$a^*, b^*, \text{RRMSE}(a^*, b^*),$

$\text{RRMSE}(1.2a^*, b^*), \text{RRMSE}(0.8a^*, b^*),$

$\text{RRMSE}(a^*, 1.2b^*), \text{RRMSE}(a^*, 0.8b^*).$

(Tip: the optimal values for α and β might be very close to extreme limits of the allowed range. Be aware of that possibility.)

(c) (6 points)

Show the following 5 images (at each pixel, show the magnitude of the pixel value; use a jet colormap) in the report using exactly the same colormap (i) Noiseless image, (ii) Noisy image,

(iii) Image denoised using quadratic prior $g_1(\cdot)$ and optimal parameter tuning;

(iv) Image denoised using Huber prior $g_1(\cdot)$ and optimal parameter tuning;

(v) Image denoised using discontinuity-adaptive prior $g_3(\cdot)$ and optimal parameter tuning.

(d) (3 points)

Show the plots of the objective-function values (vertical axis) versus iteration (horizontal axis) corresponding to each of the 3 denoised results in (iii), (iv), and (v) above.

2. **(9=6+3 points) Bayesian Denoising of a Brain Magnetic Resonance Image.**

Repeat the same task for the brain MRI image in the “data” folder.

Show the images and the plots.

3. (9=6+3 points) Bayesian Denoising of a RGB Microscopy Image.

Repeat the same task for the microscopy image in the “data” folder (assuming a Gaussian noise model as an approximation), but with the following MRF prior functions:

(A) where the penalty between two vector-valued neighboring pixels is the squared-L2-norm of their vector difference;

(B) where the penalty between two vector-valued neighboring pixels is the L2 norm of their vector difference;

(C) where the penalty between two vector-valued neighboring pixels is the (Huber-regularized) L1 norm of their vector difference.

- (a) (6 points) Show the following 5 images (at each pixel, show the magnitude of the pixel value; use a jet colormap) in the report using exactly the same colormap (i) Noiseless image, (ii) Noisy image, (iii) Image denoised using prior-A and optimal parameter tuning, (iv) Image denoised using prior-B and optimal parameter tuning, (v) Image denoised using prior-C and optimal parameter tuning.

In which of the three cases (i.e., A,B,C) do you observe edge preservation ? and why ?

- (b) (3 points) Show the plots of the objective-function values (vertical axis) versus iteration (horizontal axis) corresponding to each of the 3 denoised results above.

4. (25 points) Dictionary Learning on Image Patches, Followed by Image Denoising.

Get the 2D chest-CT image in the “data” folder.

Consider a dictionary matrix D with K columns as its atoms, where the k -th column d_k , for any k , has $\|d_k\|_2 \leq 1$ (hard constraint).

Consider the following dictionary learning formulation for some set of image patches $\{x_i\}_{i=1}^I$, under the hard constraint stated above:

$$\arg \min_D \min_r \sum_{i=1}^I \|x_i - Dr_i\|_2^2 + \lambda \|r_i\|_p^p,$$

where each r_i is a column vector (of length K) containing the coefficients used in the representation of image patch x_i , and $p > 0$ parameterizes the norm/quasi-norm of the vector r_i .

(a) (9 points)

Implement a function to learn the dictionary D for image patches of size 8×8 pixels. The dictionary D should have the number of atoms $K = 64$. Your optimization algorithm should be the same for any suitable choice of p .

For computational efficiency, the learning must only use a subset of image patches that have a sufficiently large variance of intensities within the patch (e.g., avoiding constant-intensity patches). Nevertheless, the final dictionary must indeed be able to model and fit all kinds of patches. With these design motivations, engineer a strategy to create the subset of patches $\{x_i\}_{i=1}^I$ (within the given image) that will be used to learn D , and justify your strategy.

(b) (6 points)

Use your function to learn four different dictionaries for values of $p = 2, p = 1.6, p = 1.2, p = 0.8$.

• (2 points)

In each of the 4 cases, during the optimization process, show the graph of the objective function versus iterations.

• (2 points)

In each of the 4 cases, before and after the optimization process, show the atoms used as the initial estimate (must be same for all cases) and the optimal estimates. Describe any trend that you observe.

• (2 points)

In each of the 4 cases, after the optimization process, show the histogram of coefficients within each r_i , pooled across all r_i . Describe and justify the trend that you (expect to) observe.

(c) (10 points)

• (2 points)

Given the 2D chest-CT image, simulate a noisy (or noisier) version by introducing i.i.d. Gaussian zero-mean noise of standard deviation equaling 10% of the intensity range in the given image.

• (4 points)

You want to use the learned dictionary for $p = 0.8$ to denoise the simulated noisy image (not just a subset of its patches). Propose an optimization problem for the same, and implement a function to compute an optimal estimate of the denoised image.

• (2 points)

During the optimization process, show the graph of the objective function versus iterations.

• (2 points)

Show the simulated noisy image, the denoised image, and the original/given image.